

NATHAN FERNANDO

AI & Data Engineer

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ABOUT ME

AI & Data Engineer with hands-on experience in AI and machine learning projects, turning business problems into working AI and data solutions. I build AI agents and computer vision tools in Python, using LangGraph, YOLO and TensorFlow, backed by SQL databases and REST APIs. I have also delivered fintech, payroll, inventory and logistics systems with C#/.NET. My projects include a multi-agent AI travel planner and a motorcycle detection app. I work well with customers, learn fast, and like clear, useful results.

SKILLS

Languages: Python, C#, TypeScript, JavaScript, Java, SQL

Backend & Frontend: .NET / ASP.NET Core, Angular, Node.js, HTML5, CSS3, RESTful APIs, FastAPI, React, Dapper ORM

Databases & Reporting: Microsoft SQL Server, AWS, SSRS, PostgreSQL, MySQL, MongoDB, SQLite

AI & Agents: LangGraph, LangChain, LLM APIs (Groq), multi-agent orchestration, tool/API integration, TensorFlow, CNNs, YOLO, ByteTrack, computer vision, predictive modeling

Tools & Practices: Git/GitHub, AWS fundamentals, Linux shell basics, Agile/Scrum, Docker, ERP system integration

EXTRA CURRICULAR ACTIVITIES

SIS & Leo Club, Interact Club: Community service, leadership and fundraising.

Athletics (Rugby & Swimming): Cultivated discipline through competitive sports.

Environment Society (President): Led school sustainability projects.

REFEREES

Dr Gayana Fernando
Head - School of Information Technology
SLIIT City Uni
Tel: 011 754 3600
gayana.f@scu.lk

Dr Nipunika Vithana
Senior Lecturer, Faculty of Computing and Technology
SLIIT City Uni
Tel: 011 754 3612
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WORK EXPERIENCE

SCIENTER TECHNOLOGIES (PTE) LTD - SOFTWARE ENGINEER INTERN
Jun 2026 - Nov 2026

- Worked in an Agile team building fintech apps with .NET, SQL Server and Angular.
- Built REST API connections and SQL stored procedures that turned business rules into working features, fixed bugs and helped with deployments.
- Visited customer sites to demonstrate systems and help set them up.

EDUCATION

BSc (Hons) in Computer Science | Expected June 2027
University of Bedfordshire

Higher National Diploma in Information Technology | Completed 2026
SLIIT City University (SCU) | GPA 3.5 / 4.0 | Dean's List (Year 2, Semester 2)

Foundation Certificate in IT | 2023-2024, SLIIT City Uni (SCU)

G.C.E. Ordinary Level (2022/23) | Wesley College, Colombo | 4As, 2Bs, 3Cs

CERTIFICATIONS

- **AWS Cloud Practitioner Essentials** course (2026), AWS
- **Operating Systems Basics** course (2026), Cisco Networking Academy
- **AI Generation Bootcamp** (2026), STEMLINK
- **Certificate in AI Course** (2025), SLIIT City Uni (SCU)
- **HP LIFE Professional Certifications** (2026)
- **English Diploma**, British Council

PROJECTS

TripMate AI - AI travel planner.
- Several AI agents plan flights, hotels and the trip from a plain-text request.
- **Tools:** Python, FastAPI, LangGraph, LangChain, Groq, PostgreSQL
- **Methods:** multi-agent workflow, API tool calls, saved chat state

Motorcycle Unsafe-Movement Detection - Traffic video analysis app.
- Flags suspected wrong-way and crosswalk riding in traffic camera video.
- **Tools:** Python, FastAPI, YOLO, ByteTrack, SQLite, Docker
- **Methods:** object detection and tracking, movement analysis, rule-based checks

AI Food Intelligence System - Smart recipe and nutrition app.
- Suggests healthy recipes from home ingredients, with full nutrition facts.
- **Tools:** Python, HTML
- **Methods:** two-way matching (ingredients to recipes, meals to nutrition)

Vision-Based GUI Agent- Final-year research, in progress.
- A desktop agent that reads the screen, then clicks and types to finish web tasks, and checks after each step that the action worked.
- **Tools:** Python, PyTorch, YOLO, Playwright, PyAutoGUI, PaddleOCR

ENTERPRISE PROJECTS *(Completed during internship at Scienter Technologies)*

Destinity Container Maintenance System - Container maintenance platform.
- Rebuilt the system with Angular and C# to make it faster and easier to grow. Wrote SQL Server stored procedures for live container tracking and filtering.

Destinity Payroll System - HR and payroll system.
- Automated EPF/ETF file creation in C#, replacing a manual process, and improved leave calculations. Fixed bugs so payroll results are accurate.

Client Maintenance System - Company system for managing clients and distributors.
- Built SSRS reports and SQL Server stored procedures, added Angular components, fixed bugs, and helped upgrade ASP.NET Core and Angular.

Sales Inventory & Destinity Management Systems - Inventory and ERP platforms.
- Built ERP-connected features and Angular screens for inventory, sales and logistics tracking. Wrote stored procedures and made C# API endpoints and queries faster.